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LEARNER GUIDE

For

Effective Project Management for Small Projects

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Conducted by

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
17.0	19 August 2026	Legacy master trainer slide release used as the controlling syllabus.	Tertiary Infotech Academy
18.0	19 August 2026	Complete visual and instructional redesign; content enriched from the Agile and PMP references; seven connected tool-based activities and self-contained editable template packs added; LG and LP regenerated.	Dr Alfred Ang
19.0	19 August 2026	Expanded to the complete five-phase lifecycle and 200 substantive trainer slides; twelve connected tool-based activities, six new lifecycle workbooks, deeper reference-sourced guidance and an aspect-correct cover hero added; LG and LP regenerated.	Dr Alfred Ang

TABLE OF CONTENTS

1. Course Overview

This guide accompanies the visual trainer deck. Detailed procedures live here and in the self-contained activity folders; the PowerPoint intentionally contains only activity briefings and success criteria.

1.1 Learning Outcomes

- LO1: Develop project scope and team to initiate the facilitation of a small project
- LO2: Develop a realistic project plan based on the agreed project scope
- LO3: Utilise appropriate methods and tools to track and drive a project
- LO4: Identify, analyse and manage project risks
- LO5: Collaborate effectively with project stakeholders
- LO6: Deploy manpower, budget and other resources for efficient completion
- LO7: Track project deliverables and correct misalignments through control and closeout

1.2 Skills Framework Alignment

Code	Ability / Knowledge	Evidence
A1	Facilitate execution of small projects that are typically less than six months, with limited budget and interdependencies, and no significant strategic impact	Activities 1–2
A2	Implement realistic project plans based on project objectives and scope	Activities 3–5
A3	Utilise appropriate methods and tools to track and drive progress against plans and timelines	Activities 6 and 10
A4	Identify risks to project success and take appropriate actions to manage them	Activity 7
A5	Collaborate effectively with relevant internal and external stakeholders directly impacting the project	Activity 8
A6	Deploy manpower, financial budgets and relevant resources for efficient and effective completion	Activity 9
A7	Track deliverables against schedules, monitor costs,	Activities 11–12

	timescales and resources, and take basic corrective actions when misaligned	
K1	Elements of a small project	Topic 1
K2	Requirements of a project plan	Topic 2
K3	Application of appropriate project management methodologies and tools	Topic 3
K4	Project risks	Topic 4
K5	Project stakeholder identification	Topic 5

1.3 Running Case

A 42-person Singapore specialty grocer will introduce online ordering and two-hour click-and-collect at three outlets. The five-month pilot has a S$180,000 cap, a core team of seven, two external suppliers, and limited dependencies. It is important to operations but does not change the company's overall strategy, making it a genuine small project.

Success target: Launch at all three outlets by 30 January 2027; process at least 120 pilot orders per week; keep pick accuracy at or above 98%; achieve customer satisfaction of at least 4.2/5; and remain within the approved S$180,000 budget.

1.4 Complete Project Lifecycle

Phase	Purpose
Initiation	Select, justify, authorise and align the project
Planning	Integrate requirements, scope, schedule, cost, quality, resources and risk
Execution	Coordinate work, people, suppliers, engagement and knowledge
Monitoring and Controlling	Measure, forecast, resolve, change and correct
Closure	Accept, hand over, close obligations, learn and transfer benefits

The phases overlap in practice. Integration, governance, stakeholder engagement, risk and quality continue across the lifecycle.

1.5 Tool-Based Evidence

Each activity folder includes relevant editable Office tools from the supplied source library plus purpose-built v19 lifecycle workbooks. Save completed tools with clear versioned filenames.

1.6 Supplied Reference Base

- WSQ - Master Trainer Slides - Effective Project Management for Small Projects - v17.pptx
- PMP-35-PDU-Training-Slides-v16.pptx

- WSQ - Master Trainer Slides - Agile Project Management for Business - v9.0.pptx
- Project Management templates guides.pdf

Topic 01 — Initiation of a Project

Alignment: A1, K1 · LO1

Small-project fit · value · charter · scope · team · kickoff

Why This Topic Matters

Use the small-project screen before tailoring governance. A five-month, S$180,000 pilot with seven people needs proportionate controls, but it still needs authorisation, baselines and decisions.

Core Concepts

- A project is temporary, unique and value-seeking; operations are ongoing and repetitive.
- A small project is normally under six months, has a limited budget and team, few dependencies and no significant strategic impact.
- The project manager integrates people, decisions and trade-offs; they do not merely maintain a schedule.
- Success balances value and outcomes with scope, schedule, cost, quality, resources and risk.
- Predictive, adaptive and hybrid approaches are choices driven by uncertainty, not labels of maturity.
- The business case tests whether the project should exist; the charter authorises it to exist.
- A useful charter names purpose, measurable success, scope boundaries, milestones, budget, major risks, team and sponsor authority.
- A kickoff creates one shared picture of why, what, who, how and what happens next.

Applied Guidance

Use the small-project screen before tailoring governance. A five-month, S$180,000 pilot with seven people needs proportionate controls, but it still needs authorisation, baselines and decisions.

Business value includes financial return, service improvement, risk reduction and capability. Do not combine these into one unsupported number.

A charter is strongest when it exposes what the project will not do and what the project manager may decide without returning to the sponsor.

For hybrid work, keep the charter and milestone gates while allowing short build-and-feedback cycles inside the delivery phase.

Reference-Sourced Deep Dives

Project Work versus Operations

Dimension	Project	Operations

Purpose	Create a unique result	Sustain recurring value
Duration	Temporary	Ongoing
Change	Introduces change	Stabilises and improves work
Closure	Formal endpoint	Continues until redesigned

Decision insight: The same people may do both; the management system changes with the nature of the work.

The Small-Project Qualification Test

- Time: Normally less than six months.
- Budget: Limited enough for proportionate governance.
- Team: Small core team with clear interfaces.
- Dependencies: Few and manageable.
- Impact: No significant strategic or enterprise-wide effect.
- Uncertainty: Can be bounded through short learning cycles.

From Output to Benefits

1. Output — click-and-collect capability

2. Outcome — orders collected predictably

3. Benefit — revenue, satisfaction and efficiency

4. Measure — owner, baseline, target and date

Decision insight: Deliverables are produced by the project; benefits are usually realised with operations.

Project Selection Methods

Method	Best for	Watch for
Cost-benefit	Financial viability	Unsupported benefit assumptions
Weighted scoring	Multiple criteria	Biased weights
Payback	Liquidity and speed	Ignores later value
Mandatory driver	Compliance or safety	Treating obligation as optional

Decision insight: Use more than one lens when a small financial difference hides a major operational consequence.

Sponsor and Project Manager—Different Accountability

- Sponsor: Owns investment, priority and escalation.
- Project manager: Integrates delivery and advises decisions.
- Business owner: Owns adoption and benefits after handover.
- Functional manager: Provides capability and resolves resource conflicts.

Business Documents at Initiation

Artefact	Decision supported	Owner
Business case	Invest, hold or reject	Sponsor / business owner

Benefits plan	How value will be measured	Benefits owner
Project charter	Authorise and delegate	Sponsor with PM
Assumptions log	What must be validated	Project manager

Decision insight: Keep documents brief, but never combine ownership so completely that accountability disappears.

A Charter That Can Govern

- Purpose: The problem and desired outcome.
- Success: Measures, targets and dates.
- Boundaries: Deliverables and exclusions.
- Authority: Sponsor, PM and tolerances.
- Frame: Milestones, budget and constraints.
- Exposure: Assumptions, dependencies and high risks.

Assumption, Constraint, Dependency or Risk?

Type	Meaning	Example
Assumption	Believed true for planning	Outlet Wi-Fi capacity is sufficient
Constraint	Fixed limitation	S$180,000 ceiling
Dependency	Reliance on other work	POS API access before integration
Risk	Uncertain event and impact	API access may be late

Decision insight: Classify correctly because each item needs a different response.

Lightweight Governance with Tolerances

1. Set — scope, time, cost and risk limits

2. Measure — compare current forecast

3. Act — PM decides within authority

4. Escalate — sponsor decides beyond tolerance

Decision insight: Authority without tolerances creates hesitation; tolerances without reporting create surprise.

Kickoff as an Alignment Decision

1. Why — value and success

2. What — scope and exclusions

3. Who — roles and decisions

4. How — method, rhythm and evidence

5. Next — 48-hour actions and owners

Decision insight: A good kickoff reveals mismatched assumptions early enough to repair them cheaply.

Activity 1 — Screen and Select the HarbourHub Pilot

Field	Detail
Phase	Initiation
Duration	35 minutes
Objective	Confirm that the proposed work is a viable small project and recommend the best option (A1, K1).
Tools	Small Project Screening, Multi-Criteria Selection and Cost-Benefit Analysis
Situation	HarbourHub Fresh has three competing improvement ideas: click-and-collect, a loyalty app, and a warehouse automation study. Only one can be funded this quarter. The sponsor needs a transparent selection decision and evidence that the selected work fits the small-project profile.
Output	A completed small-project screen, weighted option scorecard, cost-benefit check and sponsor-ready selection recommendation.

Tool Files

- 02 Cost Benefit Analysis.xlsx
- 01 Small Project Screening and Selection.xlsx

Detailed Procedure

1. Read the three option briefs and separate desired outcomes from proposed solutions.

2. Apply the small-project screen: temporary, unique, under six months, limited budget/team and limited interdependencies.

3. Agree five weighted decision criteria covering value, feasibility, urgency, risk and strategic fit.

4. Score each option using evidence; record the rationale beside every score.

5. Open the cost-benefit template and test the preferred option using the S$180,000 ceiling and a 20% downside case.

6. Record assumptions, exclusions and one non-financial factor that the arithmetic does not capture.

7. Recommend approve for chartering, hold for more evidence, or reject; state the sponsor decision required.

Acceptance Criteria

The selected option satisfies every small-project criterion, the weighted score reconciles, the financial downside is visible and the recommendation states a clear gate decision.

Debrief

- Selection prevents the team from efficiently delivering the wrong project.
- Weights expose priorities; narrative rationale prevents false precision.
- A selected idea is still not authorised work until the charter gate is passed.

Activity 2 — Authorise the Pilot and Run the Kickoff Gate

Field	Detail
Phase	Initiation
Duration	45 minutes
Objective	Define scope, success, authority and team arrangements for an executable small-project start (A1, K1).
Tools	Business Case on a Page and Project Charter
Situation	The click-and-collect pilot has been selected, but Operations, Finance and the two vendors hold different assumptions about success, exclusions and decision authority. The sponsor will authorise delivery only after a concise gate review.
Output	A one-page business case, signed-ready charter, assumptions and constraints log, milestone frame, decision tolerances and kickoff action list.

Tool Files

- 05 Business Case on a Page.docx
- 08 Project Charter Template.xlsx

Detailed Procedure

1. Transfer the selection rationale and downside assumptions from Activity 1 into the business case.

2. Define measurable outcomes, benefits owner and the date on which benefits will be reviewed.

3. Write in-scope deliverables and explicit exclusions; test each boundary with the sponsor scenario.

4. Set five-month milestone limits, the S$180,000 ceiling and quality/schedule tolerances.

5. Name the sponsor, project manager, core team and vendor interfaces; state the project manager's authority.

6. Record assumptions, constraints, dependencies and the top three high-level risks.

7. Run the sponsor gate: approve, approve with conditions, hold or reject; capture rationale and conditions.

8. Conduct a 15-minute kickoff covering why, what, who, how, decisions and the next 48-hour actions.

Acceptance Criteria

The charter names one accountable sponsor, measurable success, explicit exclusions, decision tolerances and an unambiguous authorisation decision within the small-project constraints.

Debrief

- The business case justifies investment; the charter grants authority.
- Decision tolerances make lightweight governance possible.
- Kickoff converts signed paper into shared commitments and immediate action.

Topic 02 — Develop a Project Plan

Alignment: A2, K2 · LO2

Requirements · scope baseline · WBS · network · schedule · budget

Why This Topic Matters

Start requirements with the stakeholder need, then write acceptance evidence. Words such as easy, fast and user-friendly are not testable until a measure or observation is defined.

Core Concepts

- The plan integrates subsidiary decisions; separate scope, schedule and cost files are not yet a project plan.
- Progressive elaboration and rolling-wave planning add detail as uncertainty reduces without abandoning control.
- Requirements must be testable, prioritised and traceable from need through deliverable and acceptance evidence.
- The scope statement defines deliverables, exclusions, constraints, assumptions and acceptance criteria.
- The WBS decomposes deliverables to manageable work packages; the WBS dictionary removes ambiguity.
- Dependencies create the network; the longest path sets the earliest finish and carries zero or least float.
- Duration estimates should separate work, waiting, risk and resource availability; use analogous, parametric, bottom-up or three-point methods deliberately.
- The cost baseline is time-phased; contingency reserve covers known-unknowns while management reserve covers unknown-unknowns.

Applied Guidance

Start requirements with the stakeholder need, then write acceptance evidence. Words such as easy, fast and user-friendly are not testable until a measure or observation is defined.

The WBS follows deliverables rather than the organisation chart. Work packages should be large enough to manage and small enough to estimate, assign and accept.

The critical path is the longest-duration path through the current network. It is not necessarily the most technically difficult or politically important work.

A baseline is the approved reference for control. A forecast is the latest expected outcome. Keep both visible; overwriting the baseline destroys the ability to explain variance.

Reference-Sourced Deep Dives

The Integrated Project Plan

- Scope: What is authorised and accepted.
- Schedule: Sequence, duration and milestones.
- Cost: Time-phased funding and reserves.
- Quality: Measures, assurance and control.
- Resources: Skills, capacity and procurement.
- Risk & stakeholders: Responses, engagement and governance.

Requirements Elicitation

1. Identify — sources and decision roles

2. Gather — interview, observe, workshop

3. Analyse — conflicts, constraints and value

4. Specify — testable requirement

5. Validate — source confirms meaning

Decision insight: Elicitation is discovery and negotiation—not taking an order from one stakeholder.

Weak versus Testable Requirements

Weak wording	Testable wording	Evidence
Fast collection	90% ready within 10 minutes	Timestamp sample
Accurate picking	At least 98% line accuracy	Check sheet
Easy to use	New staff completes task after 30-minute training	Observed test
Reliable	99.5% availability in pilot hours	System log

Decision insight: A measure without a method or status date is still ambiguous.

Prioritisation Choices

Method	Decision logic	Use when
MoSCoW	Must / Should / Could / Won't	Time-boxed scope
Weighted scoring	Value across criteria	Competing options
Kano	Basic, performance, delight	Customer experience
Trade-off sliders	Relative constraint priority	Governance alignment

Decision insight: Priorities must remain traceable to outcomes, not just stakeholder preference.

Traceability from Need to Evidence

1. Outcome — what success changes

2. Requirement — what must be true

3. Deliverable — what is produced

4. Work package — what the team performs

5. Acceptance — how completion is proven

Decision insight: Trace both directions: every need is delivered, and every deliverable has a reason.

Scope Statement Essentials

- Product scope: Characteristics of the result.
- Project scope: Work required to create it.
- Exclusions: What is deliberately not included.
- Acceptance: Who approves against which criteria.
- Constraints: Limits the plan must respect.
- Assumptions: Planning beliefs to validate.

WBS Decomposition

1. Start — authorised deliverables

2. Break down — smaller deliverable components

3. Test 100% — all work, no overlap

4. Stop — estimable, assignable, sequenced, accepted

5. Define — dictionary entry

Decision insight: Do not decompose by department when the deliverable crosses departments.

WBS Dictionary—Why It Matters

Field	Decision enabled	Failure prevented
Description	Shared meaning	Different estimates for different work
Owner	Accountability	Unclaimed delivery
Acceptance	Definition of complete	Almost done forever
Dependencies	Sequence	Late discovery
Estimate basis	Confidence and revision	Unexplained padding

Decision insight: The diagram shows structure; the dictionary makes the structure manageable.

Dependency Relationships

Type	Logic	Example
Finish-to-start	A finishes before B starts	Configure before integration test
Start-to-start	A starts before B starts	Draft training after testing

		begins
Finish-to-finish	A finishes before B finishes	Documentation completes with testing
Start-to-finish	Rare transition logic	Old support ends after new starts

Decision insight: Leads and lags change timing; they do not remove the underlying dependency.

Critical Path Calculation

1. Forward pass — earliest start and finish

2. Project finish — largest earliest finish

3. Backward pass — latest finish and start

4. Float — late start minus early start

5. Critical path — least-float path

Decision insight: The critical path is the longest-duration route through current logic, not the most important-looking task.

Estimating Methods

Method	Strength	Limitation
Analogous	Fast early estimate	Lower detail
Parametric	Repeatable unit logic	Needs valid relationship
Bottom-up	Detailed and accountable	Time-consuming
Three-point	Makes uncertainty visible	Still depends on judgement

Decision insight: Record the basis so the estimate can be challenged and updated without starting over.

Cost Baseline and Reserves

- Work-package estimates: Rolled up from authorised scope.
- Contingency reserve: Funded responses for identified risk.
- Cost baseline: Time-phased performance reference.
- Management reserve: Held outside baseline for unknown-unknowns.
- Project budget: Cost baseline plus management reserve.

Activity 3 — Discover Requirements and Map Acceptance Evidence

Field	Detail
Phase	Planning
Duration	35 minutes
Objective	Elicit, prioritise and trace requirements from stakeholder need to acceptance (A2, K2).
Tools	Requirements Workshop, Stakeholder Register and Traceability Matrix
Situation	Outlet supervisors want speed, Finance

	wants payment control, customers want predictable collection, and the POS vendor wants a stable interface. The charter is approved, but the stated needs are not testable.
Output	A stakeholder register, workshop decision log and requirements traceability matrix with priorities, owners, deliverables and measurable acceptance evidence.

Tool Files

- 25 Requirements Traceability Matrix.xlsx
- 03 Requirements Workshop and Stakeholder Register.xlsx

Detailed Procedure

1. Identify requirement sources and record their influence, impact and decision role.

2. Use interviews, observation and a facilitated workshop to surface needs, constraints and conflicts.

3. Rewrite vague words such as fast, simple and reliable into measurable conditions.

4. Give every requirement an ID, source, rationale, owner and related charter outcome.

5. Apply MoSCoW and challenge each Must against the pilot success criteria.

6. Define a deliverable and acceptance method for every Must and Should.

7. Record unresolved questions, decision owners and due dates in the workshop log.

8. Run a traceability audit and repair any orphan outcome, requirement or acceptance test.

Acceptance Criteria

Every prioritised requirement has a source, rationale, owner, deliverable and testable acceptance method; unresolved conflicts have named decision owners and dates.

Debrief

- A requirement is not complete until acceptance can be observed or measured.
- Prioritisation is a trade-off decision, not a label applied by the loudest stakeholder.
- Traceability protects value when scope changes later.

Activity 4 — Build the Scope Baseline with WBS and Dictionary

Field	Detail
Phase	Planning
Duration	45 minutes
Objective	Turn approved requirements into a complete, manageable scope baseline

	(A2, K2).
Tools	Work Breakdown Structure, WBS Dictionary and Requirements Traceability Matrix
Situation	The team has 14 prioritised requirements, but the vendor proposal groups work by department and omits outlet readiness, training and handover. Estimates cannot be trusted until the total scope is visible.
Output	A scope statement, WBS, WBS dictionary and repaired traceability matrix with no missing or duplicated work.

Tool Files

- 24 Work Breakdown Structure.pptx
- 24 WBS Dictionary.xlsx
- 25 Requirements Traceability Matrix.xlsx

Detailed Procedure

1. Write the scope statement: deliverables, exclusions, assumptions, constraints and acceptance conditions.

2. Decompose by deliverable rather than department; include project management and transition work.

3. Apply the 100% rule and test for overlap between sibling elements.

4. Stop decomposition where a work package can be estimated, assigned, sequenced and accepted.

5. Complete dictionary entries for description, owner, acceptance, dependencies, assumptions and estimate basis.

6. Map each requirement to at least one work package and each work package to a requirement or project-management need.

7. Ask another team to find omitted, duplicated or ambiguous scope and repair the baseline.

Acceptance Criteria

The WBS covers 100% of authorised work without overlap, each work package is manageable and every requirement traces to acceptance-ready scope.

Debrief

- The WBS describes all project work; the schedule describes when that work occurs.
- Transition and project-management work are real scope even when customers never see them.
- A strong dictionary prevents different teams from estimating different meanings.

Activity 5 — Develop the Integrated Schedule and Cost Baseline

Field	Detail
Phase	Planning
Duration	55 minutes
Objective	Sequence, estimate and fund the authorised scope as one realistic plan (A2, K2).
Tools	Network Diagram, Gantt Schedule and Budget versus Actual
Situation	The launch must precede Lunar New Year trading. The vendor has proposed a date without dependency logic, resource calendars or reserves. Finance requires a time-phased baseline within S$180,000.
Output	A network with forward/backward passes, critical path, Gantt milestones, basis of estimates and a time-phased cost baseline plus contingency within the authorised ceiling.

Tool Files

- 28 Schedule Network Diagram.xlsx
- 29 - Gantt Chart - Schedule - Daily Weekly Monthly and Phases.xlsx
- 32 Project Budget - Planned vs Actual - Weekly and Monthly.xlsx

Detailed Procedure

1. Translate WBS work packages into activities and identify mandatory and discretionary dependencies.

2. Add finish-to-start, start-to-start, leads and lags only where the logic is defensible.

3. Estimate durations using analogous, parametric, bottom-up or three-point methods and record the basis.

4. Perform forward and backward passes; calculate total float and identify the current critical path.

5. Build the Gantt with decision milestones, resource calendars and acceptance gates.

6. Estimate cost bottom-up and cross-check with an analogous or parametric estimate.

7. Time-phase labour, vendor, equipment, training and contingency; keep management reserve separate.

8. Reconcile every scheduled work package to the budget and every milestone to acceptance evidence.

Acceptance Criteria

The dependency network calculates correctly, the plan fits the five-month limit, all authorised work is time-phased and funded, and baseline plus contingency stays within S$180,000.

Debrief

- A Gantt is the presentation of schedule logic, not the logic itself.
- The critical path can change after every update or approved change.
- Baseline and forecast must remain separate so variance can be explained.

Topic 03 — Apply Methods and Tools to Drive a Project

Alignment: A3, K3 · LO3

Quality planning · flow · root cause · Pareto · improvement

Why This Topic Matters

Tool choice begins with the question. Use a check sheet for facts, fishbone for breadth, 5 Whys for depth, Pareto for priority and a control chart for behaviour over time.

Core Concepts

- A method defines how work is managed; a tool makes information visible. Choose both for the decision needed.
- Predictive plans optimise sequence and baseline control; adaptive boards optimise flow and feedback; hybrid combines both intentionally.
- Quality means conformance to requirements and fitness for use, not luxury or perfection.
- Prevention and appraisal are usually cheaper than internal and external failure.
- A check sheet captures facts, a control chart detects process behaviour, Pareto prioritises causes, and fishbone plus 5 Whys investigates root cause.
- Root-cause analysis must end at a process or system cause the team can influence, not blame a person.
- Visual work management exposes blocked work and excessive WIP before the schedule report does.
- Continuous improvement closes the loop through a named action, owner, measure and review date.

Applied Guidance

Tool choice begins with the question. Use a check sheet for facts, fishbone for breadth, 5 Whys for depth, Pareto for priority and a control chart for behaviour over time.

Common-cause variation belongs to the process; special-cause variation has an identifiable event or condition. Tampering with a stable process after every data point can make performance worse.

Corrective action addresses an observed variance or non-conformity. Preventive action lowers the probability of future variance. Both need an owner and verification.

Visual work management should expose blocked work, WIP, ageing and ownership—not become a colourful task list with no decision rules.

Reference-Sourced Deep Dives

Method versus Tool

Element	Purpose	Example
Method	Defines how work is	Predictive, Scrum,

	managed	Kanban, PDCA
Tool	Makes evidence visible	WBS, board, chart, log
Policy	Defines operating rule	WIP limit, acceptance rule
Cadence	Defines review rhythm	Daily review, weekly status

Decision insight: A familiar tool used without a decision rule becomes decoration.

Predictive, Adaptive and Hybrid

Signal	Predictive	Adaptive / hybrid
Requirements	Stable and knowable	Evolving or discovery-heavy
Delivery	Larger gated releases	Short usable increments
Change	Controlled against baseline	Expected inside prioritised backlog
Best fit	Low uncertainty, costly rework	Fast feedback, reversible choices

Decision insight: Tailor by deliverable when different parts of one small project carry different uncertainty.

Direct and Manage Project Work

1. Authorise — approved plan and work package

2. Coordinate — people, suppliers and interfaces

3. Produce — deliverables and work data

4. Learn — issues, knowledge and requests

5. Update — documents and forecasts

Decision insight: Execution produces both deliverables and information needed for control.

A Lean PMIS for a Small Project

- Source of truth: Named location and owner.
- Version control: Number, date, author and status.
- Decision log: What, who, when and rationale.
- Access: Right people, current artefact.
- Archive: Superseded records remain traceable.
- Security: Sensitive content has proportionate access.

Quality Planning, Assurance and Control

Discipline	Question	Evidence
Planning	What quality is required?	Metrics, methods and acceptance
Assurance	Will the process achieve it?	Review, audit, coaching
Control	Does the result conform?	Inspection, tests and charts
Improvement	How will the system	Experiment and verification

	change?	

Decision insight: Inspection finds defects; assurance and improvement reduce the chance of creating them.

HarbourHub Quality-Cost Decision

Option	Near-term effect	Decision signal
Add catalogue-data validation	S$4,000 prevention spend	Reduces rejection and re-pick risk
Test only at pilot launch	Lower spend now	Defects discovered with customers waiting
Accept manual correction	No tooling change	Recurring labour and queue growth
Decision	Fund validation before rollout	Cheaper than repeated external failure

Decision insight: The team chooses prevention because one avoided launch-week failure can repay the control.

Quality Tool Sequence

1. Check sheet — capture facts

2. Fishbone — broaden possible causes

3. Five Whys — test one chain

4. Pareto — focus the vital few

5. Control chart — understand behaviour

6. Experiment — verify improvement

Decision insight: Do not jump from anecdote directly to solution.

Common Cause versus Special Cause

Signal	Meaning	Response
Stable random variation	System behaviour	Improve the process
Point beyond limits	Potential special cause	Investigate event
Run or trend	Non-random pattern	Test changed condition
Within limits but off specification	Stable but incapable	Redesign process

Decision insight: Control limits come from process data; specification limits come from requirements.

Kanban Operating Policies

- Visualise: Show status, owner and blockers.
- Limit WIP: Stop starting; start finishing.
- Manage flow: Watch ageing and bottlenecks.
- Make policies explicit: Entry, exit and priority rules.
- Feedback: Review at a useful cadence.
- Improve: Run small measurable experiments.

PDCA Improvement Loop

1. Plan — problem, cause, hypothesis and measure

2. Do — small controlled change

3. Check — compare evidence with prediction

4. Act — standardise, adapt or stop

Decision insight: Improvement is complete only when the new practice is verified and sustained.

Manage Project Knowledge

- Explicit: Documents, data and procedures.
- Tacit: Experience and judgement shared through work.
- Create: Reviews, experiments and problem solving.
- Use: Decision support during delivery.
- Transfer: Handover and lessons for future teams.

Activity 6 — Tailor the Delivery Method and Quality Plan

Field	Detail
Phase	Execution
Duration	35 minutes
Objective	Choose proportionate methods and tools to drive flow and prevent defects (A3, K3).
Tools	Delivery Approach Canvas, Quality Plan and Kanban Board
Situation	The catalogue and POS interface need predictable milestone control, while the outlet workflow needs rapid testing and feedback. The team must choose a hybrid delivery rhythm and define quality before build begins.
Output	A documented hybrid approach, phase and iteration rhythm, quality management plan, Definition of Done, Kanban policy and measurement plan.

Tool Files

- 30 Kanban Board.xlsx
- 35 Check Sheets.xlsx
- 34 Control Chart.xlsx
- 06 Delivery Approach and Quality Plan.xlsx

Detailed Procedure

1. Rate requirement uncertainty, technical uncertainty, feedback speed, compliance and cost of change.

2. Choose predictive, adaptive or hybrid treatment for each major deliverable and justify the boundary.

3. Define milestone gates and short test-feedback cycles inside the execution phase.

4. Write a Definition of Done that includes acceptance, evidence, documentation and handoff conditions.

5. Set quality metrics for ready-on-time, pick accuracy, defect escape and rework.

6. Design the check-sheet sampling plan and the future control-chart review rule.

7. Configure the Kanban board, entry/exit policies, WIP limits and blocker escalation threshold.

8. Confirm which artefact is the source of truth and how versions will be controlled.

Acceptance Criteria

The approach matches uncertainty, quality is measurable before work begins, WIP and evidence rules are explicit, and predictive/adaptive controls do not contradict one another.

Debrief

- Hybrid is intentional tailoring, not running two incompatible systems in parallel.
- Quality must be designed into the workflow before it is inspected at the end.
- A tool earns its place only when it changes a decision or behaviour.

Activity 10 — Diagnose the Pilot Quality Failure

Field	Detail
Phase	Monitoring and Controlling
Duration	40 minutes
Objective	Use evidence and root-cause tools to select a controlled corrective experiment (A3, K3).
Tools	Check Sheet, Fishbone, Five Whys, Pareto and Control Chart
Situation	During testing, only 82% of orders are ready on time and pick accuracy is 94%. The vendor blames store staff; store staff blame inventory updates. The sponsor wants action before facts and hypotheses are separated.
Output	A check sheet, fishbone, five-whys chain, Pareto chart, control-chart interpretation and SMART corrective experiment.

Tool Files

- 35 Check Sheets.xlsx
- 37 Ishikawa Diagram.xlsx

- 38 Five Whys.xlsx
- 36 Pareto Principle Template.xlsx
- 34 Control Chart.xlsx

Detailed Procedure

1. Enter and reconcile the supplied defect data by type, outlet, shift and date.

2. Build a fishbone across People, Process, Technology, Data, Measurement and Environment; label facts and hypotheses.

3. Select a high-frequency cause family and complete the five-whys chain to an influenceable system cause.

4. Read the chain backwards with because and repair causal jumps.

5. Rank defects in the Pareto template, calculate cumulative percentage and identify the vital few.

6. Plot daily performance and distinguish control behaviour from the customer specification.

7. Create one SMART corrective experiment with owner, measure, due date and review rule.

Acceptance Criteria

Data totals reconcile, Pareto reaches 100%, the causal chain is actionable, the control interpretation matches the signal and the experiment has an owner and verification date.

Debrief

- Fishbone broadens, Five Whys deepens and Pareto prioritises.
- Control limits describe process behaviour; specifications describe customer need.
- Improvement is project work only when funded, owned and verified.

Topic 04 — Manage Project Risk

Alignment: A4, K4 · LO4

Identify · assess · respond · contingency · monitor

Why This Topic Matters

A risk statement separates cause, uncertain event and impact. This makes the response design testable: act on the cause, reduce probability, reduce impact or prepare contingency.

Core Concepts

- Risk is an uncertain event or condition with a positive or negative effect; an issue has already happened.
- Write risks as cause-event-impact statements so the response targets the cause or reduces the impact.
- The risk register records owner, probability, impact, exposure, triggers, response, contingency and residual risk.
- Qualitative analysis prioritises quickly; quantitative analysis estimates the range of outcomes and decision value.
- Threat responses are avoid, mitigate, transfer, accept or escalate; opportunity responses are exploit, enhance, share, accept or escalate.
- A contingency plan is triggered by an observable threshold and funded from contingency reserve.
- Decision trees compare expected monetary value; tornado charts reveal which assumptions deserve the most attention.
- Risk reviews are recurring decisions, not a register created once and archived.

Applied Guidance

A risk statement separates cause, uncertain event and impact. This makes the response design testable: act on the cause, reduce probability, reduce impact or prepare contingency.

Probability-impact scores are ordinal prioritisation aids. Do not pretend that multiplying two subjective ratings creates a precise financial forecast.

Expected monetary value helps compare repeated or material choices. It can still miss reputation, safety, legal duty and strategic option value.

Residual risk remains after the response; secondary risk is introduced by the response. Both return to the register with owners.

Reference-Sourced Deep Dives

Risk, Issue and Assumption

Item	Time orientation	Management action

Risk	May happen	Owner, trigger and response
Issue	Has happened	Contain, resolve and escalate
Assumption	Believed true	Validate by a date
Constraint	Known limit	Plan within or change formally

Decision insight: A triggered risk becomes an issue; its response owner moves from readiness to action.

The Risk Management Cycle

1. Plan — scales, thresholds and roles

2. Identify — threats and opportunities

3. Analyse — priority and decision value

4. Respond — action, owner, trigger and reserve

5. Monitor — residual, secondary and emerging risk

Decision insight: Risk management repeats whenever scope, schedule, suppliers or the environment change.

Risk Breakdown Structure

- Technical: Design, interface, data and quality.
- External: Regulation, market, weather and supplier.
- Organisational: Priority, funding, skills and governance.
- Project management: Estimate, dependency, communication and scope.
- Opportunity: Faster adoption, reuse, cost saving and capability.

Writing an Actionable Risk

1. Cause — because inventory data refresh is delayed

2. Event — stock availability may be wrong

3. Impact — orders may be rejected or re-picked

4. Response — reduce cause or impact

5. Trigger — refresh latency exceeds threshold

Decision insight: Cause-event-impact wording prevents vague risks such as 'vendor risk'.

Qualitative Risk Factors

Factor	Question	Use
Probability	How likely?	Exposure priority
Impact	How severe?	Tolerance and reserve

Urgency	How soon must we act?	Response timing
Proximity	How soon may it occur?	Monitoring cadence
Dormancy	How long before impact appears?	Early detection

Decision insight: Two risks with the same score may require different action timing.

HarbourHub Threat-Response Workshop

Named threat	Chosen response	Owned action and trigger
Inventory feed is late	Mitigate	IT lead adds latency alert; trigger >15 minutes
Vendor misses integration date	Transfer + mitigate	Milestone clause plus weekly interface test
Outlet rejects new process	Mitigate	Ops lead pilots job aid and observes first 20 orders
Cyber approval exceeds tolerance	Escalate	Sponsor decides scope trade by gate date
Residual launch disruption	Accept actively	Fund two-day floor support contingency

Decision insight: Each response is specific enough to schedule, fund, monitor and verify.

Opportunity Response Strategies

Strategy	Intent	Example
Exploit	Ensure opportunity occurs	Assign best resources
Enhance	Increase probability or benefit	Accelerate pilot learning
Share	Partner for capability	Revenue-sharing supplier
Accept	Take benefit if it occurs	No proactive spend
Escalate	Move to programme or strategy	Enterprise reuse

Decision insight: Positive uncertainty deserves ownership and deliberate action too.

Response Architecture

1. Preventive — act before trigger

2. Trigger — observable threshold

3. Contingency — act when triggered

4. Fallback — act if contingency fails

5. Residual — remaining exposure

6. Secondary — new risk from response

Decision insight: A response is not operational until timing, ownership and funding are explicit.

Decision Tree and Sensitivity

Tool	Question answered	Caution
Decision tree	Which option has better	Average value can hide tail

	expected value?	consequences
Tornado chart	Which assumption drives outcome most?	Range quality determines insight
Probability-impact matrix	What needs attention first?	Ordinal scores are not precise money

Decision insight: Use more analysis only when the decision value exceeds the analysis effort.

Risk Review Agenda

- Triggered: Promote to issue and execute response.
- Changed: Re-score probability, impact and timing.
- New: Add emerging and secondary risk.
- Obsolete: Close with rationale.
- Reserve: Reconcile use and remaining exposure.
- Escalation: Decide beyond tolerance.

Activity 7 — Build the Risk Register and Fund Responses

Field	Detail
Phase	Planning and Execution
Duration	45 minutes
Objective	Identify, analyse and manage uncertainty with proportionate, funded action (A4, K4).
Tools	Risk Matrix, Decision Tree and Tornado Analysis
Situation	The POS interface may be late, refrigerated stock counts are unreliable, a key trainer has a short availability window and a faster payment option could improve adoption. Contingency reserve is S$18,000.
Output	A prioritised risk register, response and trigger plan, EMV recommendation, tornado sensitivity reading and reconciled S$18,000 contingency allocation.

Tool Files

- 48 Risk Assessment Matrix Two pages.xlsx
- 45 Decision Tree Analysis TDIDT.xlsx
- 49 Tornado Chart - Sensitivity Analysis.xlsx

Detailed Procedure

1. Separate uncertain risks from current issues and assumptions requiring validation.

2. Write at least eight risks as cause-event-impact statements and classify them in a risk breakdown structure.

3. Score probability, impact, urgency and proximity; justify the priority threshold.

4. Assign risk owners, response owners and observable triggers.

5. Choose threat and opportunity responses; define preventive, contingency and fallback actions.

6. Use the decision tree to compare payment options and state a non-financial factor outside EMV.

7. Use the tornado chart to identify the two assumptions driving the largest outcome variation.

8. Allocate no more than S$18,000 to named risks and record residual and secondary risk.

Acceptance Criteria

Priority risks have owners, triggers and funded responses; the EMV and sensitivity logic reconcile; contingency is allocated to named uncertainty and residual risk remains visible.

Debrief

- Risk management creates options before triggers become expensive issues.
- Quantitative analysis sharpens material decisions but does not replace judgement.
- Acceptance means informed monitoring and contingency, not forgetting the risk.

Topic 05 — Collaborate with Stakeholders

Alignment: A5, K5 · LO5

Identify · analyse · communicate · engage · allocate accountability

Why This Topic Matters

Stakeholder identification is continuous. Procurement, design choices, issues and change requests can create new stakeholders or alter influence and urgency.

Core Concepts

- A stakeholder can affect, be affected by, or perceive themselves to be affected by the project.
- The stakeholder register captures interest, influence, expectations, impact, communication needs and engagement strategy.
- Power-interest mapping guides attention; the salience model adds legitimacy and urgency.
- The engagement assessment matrix compares current and desired states so the gap becomes actionable.
- RACI clarifies who is Responsible, Accountable, Consulted and Informed; every deliverable needs one clear Accountable owner.
- Communication planning matches audience, message, channel, frequency, owner and feedback route.
- Engagement is two-way sense-making and decision participation, not one-way broadcasting.
- Escalation should state the decision needed, options, impact of delay and recommendation.

Applied Guidance

Stakeholder identification is continuous. Procurement, design choices, issues and change requests can create new stakeholders or alter influence and urgency.

Power-interest mapping guides management attention; salience adds legitimacy and urgency. Use both when formal authority does not explain actual influence.

A RACI matrix is a diagnostic for decision rights. One Accountable owner is the default; exceptions need an explicit governance reason.

A communication plan specifies message, audience, channel, frequency and owner. An engagement plan adds the feedback route and decision the stakeholder can shape.

Reference-Sourced Deep Dives

The Stakeholder Engagement Cycle

1. Identify — who affects or is affected

2. Analyse — interest, influence, impact and expectations

3. Plan — desired state and decision participation

4. Engage — communicate, listen and negotiate

5. Monitor — feedback, sentiment and changed influence

Decision insight: Stakeholder identification is continuous because project decisions change who is affected.

Stakeholder Register Fields

- Identity: Role, organisation and relationship.
- Interest: What outcome matters.
- Influence: Formal and informal ability to affect work.
- Impact: How the project changes their work.
- Expectations: Definition of success and concern.
- Strategy: Engagement action and owner.

Stakeholder Mapping Lenses

Lens	Dimensions	Best use
Power-interest	Authority and attention	Management strategy
Impact-influence	Effect and ability to act	Change and adoption
Power-influence	Authority and active influence	Governance
Stakeholder cube	Three dimensions	Complex populations
Salience	Power, legitimacy, urgency	Competing claims

Decision insight: A map is a hypothesis to guide action, not a permanent label.

Seven Salience Classes

- Dormant: Power only.
- Discretionary: Legitimacy only.
- Demanding: Urgency only.
- Dominant: Power and legitimacy.
- Dangerous: Power and urgency.
- Dependent: Legitimacy and urgency.
- Definitive: Power, legitimacy and urgency.

Directions of Influence

Direction	Typical stakeholders	Leadership move
Upward	Sponsor and executives	Decision brief and escalation
Downward	Team and specialists	Clarity, coaching and feedback
Sideward	Peers and functional managers	Negotiation and coordination
Outward	Customers, vendors, regulators	Expectation and interface management

Decision insight: Influence without authority depends on evidence, trust and a clear decision request.

Communication Model

1. Encode — sender frames meaning
2. Transmit — channel carries message
3. Decode — receiver interprets
4. Feedback — understanding is tested
5. Noise — language, culture, timing and medium interfere

Decision insight: Delivery is not communication success; shared understanding is.

Communication Methods

Method	Best use	Risk
Interactive	Decision and ambiguity	Time and facilitation
Push	Targeted distribution	Receipt is not understanding
Pull	Large reference set	Stakeholder may not retrieve it
Visual radiator	Shared current state	Stale information destroys trust

Decision insight: Choose channel based on urgency, complexity, sensitivity and need for feedback.

Stakeholder Engagement Levels

Level	Signal	Management aim
Unaware	Does not know project	Build awareness
Resistant	Opposes change	Understand concern and involve
Neutral	Neither resists nor supports	Connect to relevant value
Supportive	Helps when asked	Sustain participation
Leading	Actively champions	Enable influence

Decision insight: The gap between current and desired state must become an owned action.

HarbourHub RACI Failure Test

Observed row	Failure signal	Repair
Pilot go-live	Sponsor and PM both marked A	Sponsor is sole A; PM remains R
Inventory mapping	Vendor named R with no internal owner	IT lead becomes A; vendor remains R
Outlet job aid	No R assigned	Operations lead owns production
Customer message	Six people marked C	Consult legal and operations only

Benefits review	Everyone marked I	Name finance owner and decision audience

Decision insight: The repaired matrix makes every critical deliverable and decision executable.

Evidence-Based Escalation

1. Decision — what is needed

2. Options — feasible choices

3. Impact — consequence of delay

4. Recommendation — preferred option and why

5. Deadline — latest useful decision time

Decision insight: Escalation transfers a decision, not a problem without analysis.

Activity 8 — Align Stakeholders, Accountability and Communications

Field	Detail
Phase	Execution
Duration	40 minutes
Objective	Collaborate with stakeholders and clarify decision rights for execution (A5, K5).
Tools	Stakeholder Map, Salience Model, RACI and Engagement Assessment
Situation	The POS vendor and Operations each believe the other owns interface acceptance. Finance has not approved a payment milestone, store managers receive conflicting instructions and one influential supervisor resists the pilot.
Output	A stakeholder map, salience reading, validated RACI, engagement assessment, communication actions and one-minute escalation recommendation.

Tool Files

- 42 Stakeholder Impact Over Influence Matrix.xlsx
- 43 Salience Chart.pptx
- 39 RACI Template.xlsx
- 41 Stakeholder Engagement Assessment Matrix.xlsx

Detailed Procedure

1. Identify internal, external, upward, downward and sideward stakeholders.

2. Position stakeholders on impact/influence and justify every high rating with evidence.

3. Apply the salience lens of power, legitimacy and urgency; compare it with the two-dimensional map.

4. Build RACI rows around deliverables and decisions; enforce one Accountable owner per critical row.

5. Correct missing R/A, duplicated A and excessive consultation patterns.

6. Rate current and desired engagement and define the decision each stakeholder must help shape.

7. Set message, channel, frequency, owner and feedback route for each material gap.

8. Deliver an escalation brief: decision, options, impact of delay, recommendation and deadline.

Acceptance Criteria

Critical decisions have one Accountable owner, material engagement gaps have two-way actions and the escalation asks the authorised person for a clear, time-bound decision.

Debrief

- Many communication failures begin as unclear decision rights.
- Engagement is participation and feedback, not message broadcasting.
- Attention follows influence, impact and urgency—not title alone.

Topic 06 — Deploy Resources

Alignment: A6 · LO6

People · budget · equipment · procurement · team performance

Why This Topic Matters

Resource availability is the fraction of time usable for this project after operations, leave and other commitments—not the employee's full calendar.

Core Concepts

- Resource planning links each work package to required skills, quantities, timing and availability.
- A responsibility assignment is not a capacity plan; availability must be tested against concurrent work.
- Resource levelling changes dates to resolve over-allocation; smoothing stays within available float.
- Fast tracking overlaps work and raises risk; crashing adds cost to critical-path work and may not shorten the project.
- Procurement decisions consider make-or-buy, lead time, quality, interfaces, contract type and vendor capability.
- Tuckman's stages require adaptive leadership: direct, coach, support and delegate as the team matures.
- Conflict is normal; collaborate for important relationships and outcomes, compromise when time is limited, and escalate only with evidence.
- A resource dashboard should show demand, capacity, variance, bottlenecks and decisions—not just headcount.

Applied Guidance

Resource availability is the fraction of time usable for this project after operations, leave and other commitments—not the employee's full calendar.

Smoothing uses available float and protects the end date. Levelling resolves over-allocation even when the project finish must move.

Crashing adds cost to critical-path work; fast tracking overlaps work that was sequential. Recalculate the network and risk after either action.

Conflict mode should match outcome importance and relationship importance. Collaboration is powerful but costly; not every low-stakes disagreement needs it.

Reference-Sourced Deep Dives

Resource Management Plan

- Identify: People, equipment, facilities and funding.
- Estimate: Quantity, skill and timing.

- Acquire: Internal assignment or procurement.
- Develop: Capability, trust and working agreements.
- Manage: Performance, conflict and feedback.
- Control: Usage, availability and corrective action.

Organisation Structure and PM Authority

Structure	PM authority	Resource pattern
Functional	Low	Functional manager controls
Weak matrix	Limited	Coordinator role
Balanced matrix	Shared	Negotiated authority
Strong matrix	Moderate to high	Dedicated PM authority
Project-oriented	High	Dedicated team

Decision insight: Even a small project needs explicit escalation when formal authority is limited.

Role, Authority, Responsibility and Competence

Element	Question	Evidence
Role	What function is performed?	Role description
Authority	What decisions may be made?	Charter and tolerance
Responsibility	What outcome or work is owned?	RACI and work package
Competence	Can the person perform it?	Skill and experience

Decision insight: Assigning work without authority or competence creates hidden dependency.

Two-Week Capacity Worked Example

Skill	Demand	Usable capacity / decision
Integration developer	96 hours	64 hours — defer reporting interface
Outlet trainer	40 hours	48 hours — retain buffer
Data analyst	52 hours	32 hours — buy 20 specialist hours
Test lead	60 hours	56 hours — smooth four hours into next week

Decision insight: The constraint is skill-specific usable time, not the team's nominal headcount.

Acquire, Develop or Procure

Choice	Use when	Trade-off
Internal assignment	Capability exists	Priority conflict
Develop skill	Need is recurring	Learning time
Contract resource	Short-term specialist gap	Cost and knowledge transfer
Outsource deliverable	Clear result and interface	Control and vendor dependency

Decision insight: Choose the smallest intervention that resolves the critical constraint.

Make-or-Buy Decision

Criterion	Make	Buy
Capability	Builds internal skill	Accesses specialist skill
Lead time	Depends on availability	Depends on procurement and onboarding
Control	Direct management	Contract and interface control
Cost	Fixed or opportunity cost	Price plus vendor management
Knowledge	Retained internally	Requires transfer plan

Decision insight: Lowest price is not lowest total project cost when delay and interface risk are included.

Contract Type and Risk

Type	Cost risk	Best fit
Firm fixed price	More seller risk	Stable, clear scope
Cost reimbursable	More buyer risk	Uncertain scope with control
Time and materials	Shared / capped	Short-term flexible effort
Incentive arrangement	Shared performance target	Measurable outcome

Decision insight: The contract allocates financial risk; it does not eliminate delivery accountability.

HarbourHub Team-Stage Diagnostic

Observed behaviour	Stage diagnosis	Next leadership move
Vendor and IT dispute interface ownership	Storming	Facilitate roles and decision rule
Outlet champions share one job aid	Norming	Reinforce standard and peer coaching
Test team resolves defects without escalation	Performing	Delegate within tolerance
Pilot staff worry about release	Adjourning	Recognise contribution and confirm transition

Decision insight: Leadership changes with team evidence; it is not a fixed personal style.

Conflict Approaches

Approach	When useful	Trade-off
Collaborate	Outcome and relationship both matter	Time and openness
Compromise	Time-limited middle ground	Partial satisfaction
Accommodate	Relationship or learning matters more	Your need deferred
Direct / force	Emergency or authority decision	Commitment risk
Avoid	Low value or cooling period	Issue may persist

Decision insight: Conflict is information; choose the response based on stakes and relationship.

HarbourHub Bottleneck Options

Option tested	Schedule effect	Decision
Smooth outlet training	Uses two days of float	Accept — no milestone impact
Level integration work	Moves launch by one week	Reject unless scope is traded
Crash data validation	Recovers three days for S$3,600	Approve inside reserve
Fast-track testing	May recover four days	Reject — high rework and quality risk

Decision insight: The selected combination protects launch without hiding cost or quality trade-offs.

Team Performance Signals

- Delivery: Commitments met with acceptance evidence.
- Flow: WIP, blockers and ageing visible.
- Quality: Defects and rework trend.
- Collaboration: Handoffs, feedback and conflict resolved.
- Learning: Capability and knowledge transfer.
- Sustainability: Workload remains safe and realistic.

Activity 9 — Deploy Resources and Resolve the Bottleneck

Field	Detail
Phase	Execution
Duration	45 minutes
Objective	Match skills, capacity, procurement and budget to critical work (A6).
Tools	Organisation Chart, Resource Capacity, Procurement Decision, Kanban and Budget
Situation	Testing needs 22 person-days in two weeks, but only 16 are available. The business analyst is assigned to testing, training and vendor clarification. The sponsor wants evidence before accepting delay or extra cost.
Output	An organisation map, capacity heatmap, constraint-aware Kanban, procurement comparison, compression options, budget impact and sponsor recommendation.

Tool Files

- 40 Organisation Chart.pptx
- 30 Kanban Board.xlsx
- 32 Project Budget - Planned vs Actual - Weekly and Monthly.xlsx
- 09 Resource Capacity and Procurement Decision.xlsx

Detailed Procedure

1. Map sponsor, project manager, functional managers, core team and vendor interfaces.

2. Calculate usable capacity by person, skill and week after operations and other commitments.

3. Match work-package demand to capacity and quantify the constraint and queue.

4. Set WIP limits and pull rules around the constrained skill.

5. Test smoothing within float and levelling beyond available float.

6. Test crashing and fast tracking only on the current critical path; record cost and new risk.

7. Compare make/buy options using cost, lead time, quality, interface and vendor capability.

8. Update the budget and recommend an option with schedule, cost, quality and risk effects.

Acceptance Criteria

Capacity reconciles to named work and skills, the bottleneck is explicit, procurement/compression options show complete trade-offs and the chosen deployment remains within authorised constraints.

Debrief

- A RACI assignment does not create capacity.
- Manage the constraint before adding capacity elsewhere.
- Crashing buys time with cost; fast tracking buys time with risk.

Topic 07 — Track Project Deliverables and Correct Misalignments

Alignment: A7 · LO7

Performance · earned value · change · corrective action · handover · closure

Why This Topic Matters

PV, EV and AC must share the same status date and scope basis. Otherwise SPI and CPI compare unlike quantities.

Core Concepts

- Work performance data becomes information only after comparison with the approved baseline and thresholds.
- A status report should answer: where are we, why, what is forecast, what decision is needed, and who owns the action.
- Earned value integrates scope, schedule and cost: PV is planned work, EV is completed value, and AC is actual spend.
- CPI = EV/AC and SPI = EV/PV; indices below 1.00 signal inefficiency against the baseline.
- A change request states reason, impact, options and recommendation; approved changes update baselines and traceability.
- Corrective action brings performance back to plan; preventive action reduces the probability of future variance.
- Schedule compression is applied to the current critical path and re-analysed after every change.
- Closure confirms acceptance, transition, contract completion, records, benefits ownership and lessons learned.

Applied Guidance

PV, EV and AC must share the same status date and scope basis. Otherwise SPI and CPI compare unlike quantities.

CPI and SPI below 1.00 are adverse. Translate indices into plain-language forecast and decision consequences rather than reporting decimals alone.

Change control is not change prevention. It makes trade-offs explicit, records authority and updates every affected baseline after approval.

Closure includes acceptance, transition, contracts, records, lessons and benefits ownership. The project may finish while benefits continue under operations.

Reference-Sourced Deep Dives

The Monitoring and Control Cycle

1. Collect — work performance data

2. Compare — baseline, threshold and acceptance

3. Analyse — cause and forecast

4. Decide — action, change or escalation

5. Update — owners, plans and records

6. Verify — did action change outcome?

Decision insight: Control is forward-looking: the purpose of variance is to improve the forecast and decision.

Data, Information and Reports

Level	Example	Use
Data	7 defects, S$105k actual cost	Raw observation
Information	Defects above threshold, CPI 0.86	Meaning through comparison
Report	Forecast, cause, action and decision	Governance and coordination

Decision insight: A dashboard without interpretation is data display, not project control.

A Decision-Ready Status Report

- Where: Current milestone and acceptance state.
- Variance: Scope, time, cost, quality and resource.
- Why: Cause, not just symptom.
- Forecast: Expected finish and cost.
- Action: Owner, due date and verification.
- Decision: Escalation with options and deadline.

Validate Scope versus Control Scope

Process	Question	Evidence
Validate scope	Will the authorised stakeholder accept it?	Accepted deliverable
Control scope	Has authorised scope changed?	Baseline, RTM and change log
Control quality	Does the result conform?	Inspection and test record

Decision insight: Verified quality is not the same as formal customer acceptance.

Earned Value Building Blocks

- PV: Budgeted value of work planned by status date.
- EV: Budgeted value of work actually completed.
- AC: Actual cost of completed work.
- BAC: Total authorised budget for measured scope.
- Status date: The same cut-off for every measure.

Variance and Performance Indices

Measure	Formula	Interpretation
SV	EV − PV	Negative means less value completed than planned
CV	EV − AC	Negative means completed value cost more than budgeted
SPI	EV ÷ PV	Below 1.00 means schedule inefficiency
CPI	EV ÷ AC	Below 1.00 means cost inefficiency

Decision insight: Translate every index into operational consequence and required decision.

Forecasting with EAC

Situation	Formula	Assumption
Current cost efficiency continues	EAC = BAC ÷ CPI	Past cost performance persists
Remaining work re-estimated	EAC = AC + bottom-up ETC	New detailed forecast is credible
Original variance is one-off	EAC = AC + (BAC − EV)	Future returns to plan
Cost and schedule both constrain	EAC = AC + (BAC − EV)/(CPI × SPI)	Both inefficiencies persist

Decision insight: Forecast choice is a management assumption and must be stated.

Issue Management

1. Log — fact, impact and detection date

2. Contain — protect customer or milestone

3. Own — resolution and decision owner

4. Resolve — root cause and action

5. Escalate — outside authority or tolerance

6. Verify — evidence and closure

Decision insight: Do not keep an issue disguised as a risk after it has occurred.

Loyalty-Points Change Worked Example

Analysis lens	Finding	Implication
Value	Could lift repeat purchase	Benefit is plausible but unvalidated
Scope and schedule	Adds API and two test cycles	Launch slips nine working days
Cost and resource	S$18,000 plus scarce developer	Exceeds PM tolerance
Risk and quality	New data/privacy interface	Needs security and regression evidence

Decision	Defer to post-pilot release	Protect baseline; retain request and rationale

Decision insight: Useful ideas are not automatically urgent: the sponsor protects the pilot while preserving the option.

Three Types of Action

Action	Purpose	Example
Corrective	Bring future performance back to plan	Add test capacity to recover milestone
Preventive	Reduce probability of future variance	Add interface review before integration
Defect repair	Modify non-conforming deliverable	Correct catalogue mapping

Decision insight: Each action needs authority, ownership, due date and verification.

Control Across Knowledge Areas

- Schedule: Critical path, float and milestone forecast.
- Cost: Actuals, earned value and forecast.
- Quality: Conformance, variation and corrective action.
- Resources: Utilisation, capacity and bottlenecks.
- Risk: Triggers, exposure and reserve.
- Communication & procurement: Effectiveness, obligations and disputes.

Closure from Acceptance to Benefits

1. Accept — criteria and authorised sign-off

2. Transition — operations, support, data and training

3. Close contracts — claims, invoices, warranties and records

4. Learn — causes and reusable recommendations

5. Transfer benefits — owner, baseline, target and review

6. Close — archive and release resources

Decision insight: The project can close only after unresolved work is completed or formally transferred.

Activity 11 — Control Performance, Issues and Change

Field	Detail
Phase	Monitoring and Controlling
Duration	45 minutes
Objective	Track integrated performance and take authorised corrective action (A7).
Tools	Integrated Status, Change and Issue Log, Burndown, Budget and Control Chart
Situation	At week 12, PV is S$108,000, EV is S$90,000 and AC is S$105,000. The

	sponsor requests loyalty points for S$24,000 and three weeks while a vendor interface issue threatens acceptance.
Output	An integrated status report, EVM calculations and forecast, issue decision, change-impact analysis, decision log, corrective action and updated control records.

Tool Files

- 31 Burndown Chart.xlsx
- 32 Project Budget - Planned vs Actual - Weekly and Monthly.xlsx
- 34 Control Chart.xlsx
- 11 Integrated Status Change and Issue Log.xlsx

Detailed Procedure

1. Set one status date and reconcile deliverable, milestone, budget, quality, risk and resource evidence.

2. Calculate SV, CV, SPI and CPI from PV, EV and AC; interpret each in plain language.

3. Using BAC S$162,000, calculate EAC and ETC under current cost performance and compare with the ceiling.

4. Triage the interface issue by severity, owner, containment, root cause and resolution date.

5. Document the loyalty-points request and analyse scope, time, cost, quality, resource and risk impacts.

6. Recommend approve, reject, defer or trade scope; record approver, rationale and deadline.

7. Update affected baselines or the decision log and notify owners and stakeholders.

8. Choose corrective or preventive action with owner, due date and verification measure.

Acceptance Criteria

EVM and forecast reconcile, issue and change records are distinct, the decision is authorised, all affected records are updated and corrective action is measurable.

Debrief

- Status reporting is a decision system, not a colour-coded history lesson.
- Change control enables deliberate trade-offs; it does not prevent change.
- An approved change is incomplete until every affected baseline is updated.

Activity 12 — Accept, Handover and Close the Project

Field	Detail

Phase	Closure
Duration	35 minutes
Objective	Close only when deliverables, transition, contracts, records and benefits ownership are evidenced (A7).
Tools	Acceptance, Handover, Closure and Benefits Tracker plus Retrospective
Situation	All three outlets are live, but final payment is requested before complete acceptance evidence, support ownership, training records and unresolved-item decisions are consolidated. Benefits will continue after project closure.
Output	Signed acceptance matrix, handover and training evidence, unresolved-item disposition, contract closeout, lessons register, benefits tracker and formal closure recommendation.

Tool Files

- 44 Agile Retrospective.xlsx
- 12 Acceptance Handover Closure and Benefits.xlsx

Detailed Procedure

1. Trace each accepted deliverable to its requirement, criterion, evidence and authorised approver.

2. List operational ownership, support levels, data ownership, monitoring and escalation after handover.

3. Verify training completion, knowledge transfer and access to current operating documentation.

4. Resolve or formally transfer every open issue, defect, risk, change and supplier obligation.

5. Confirm final invoices, warranties, contract records and payment approval conditions.

6. Run the retrospective and convert lessons into reusable recommendations with owners.

7. Assign benefits owners, measures, baselines, targets and review dates after closure.

8. Recommend close, close with transferred actions, or remain open; record sponsor approval.

Acceptance Criteria

Acceptance is traceable, handover ownership is explicit, open items have formal disposition, contracts and records reconcile, and benefits continue under named operational owners.

Debrief

- Elapsed time does not close a project; complete evidence and transferred accountability do.
- Lessons become assets only when future owners can find and use them.
- Project work ends before benefits measurement ends.

Assessment Readiness

- Written Assessment (Case Study) — apply project management judgement to a small-project scenario.
- Oral Questioning — explain and justify decisions against the course methods and evidence.
- Open book: course slides, Learner Guide and approved course materials only.
- Answer structure: method, scenario evidence, result, trade-off and next action.

Glossary

Term	Definition
Acceptance criteria	Measurable conditions a deliverable must satisfy to be accepted.
Baseline	The approved scope, schedule or cost reference used to measure performance.
Business case	The justification comparing options, costs, benefits and risks before investment.
Charter	The document that authorises the project and grants the project manager authority.
Contingency reserve	Budget or time for identified risks whose responses are planned.
Corrective action	Action intended to bring future performance back in line with the plan.
Critical path	The longest-duration path through the schedule network, normally with least float.
Earned value	The budgeted value of work actually completed at the status date.
Float	Time an activity can move without affecting a defined schedule date.
Issue	A condition or event that has already occurred and requires action.
Milestone	A zero-duration marker for a significant event, approval or completion.
RACI	Responsible, Accountable, Consulted and Informed role mapping.
Residual risk	Risk remaining after the planned response is applied.
Risk trigger	Observable threshold or event indicating a risk is becoming imminent or has occurred.
Rolling-wave planning	Planning near-term work in detail while keeping later work at a higher level.
Scope creep	Uncontrolled addition of scope without approved time, cost or resource change.
WBS	Deliverable-oriented hierarchical decomposition of the total project scope.